MODULE 15 · DOMAIN 2 - IMPLEMENT GENERATIVE AI AND AGENTIC SOLUTIONS

Evaluating models and apps

Fabrication, relevance, quality and safety; similarity metrics, agent evaluators, custom evaluators, data sources and evaluation levels.

DOMAIN 2 - Generative AI and agents WEIGHT 30-35% MODULE 15 of 39

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Evaluation is the **MOT** before the car goes on the road. Choose each test by the **failure you fear**, not by what's on the list.

The failure you fear, and the evaluators that catch it

FAILURE	EVALUATOR FAMILY
Invented facts	Groundedness, Groundedness Pro, Ungrounded Attributes
Off-topic or unhelpful answers	Relevance, Response Completeness
Poorly written answers	Coherence, Fluency
Wrong compared with a known answer	Similarity, F1, BLEU, ROUGE, METEOR
Harmful or unsafe output	Risk and safety evaluators
Agents doing the wrong thing	Agent evaluators: tools, tasks, intent

ON THE EXAM

Your own rules, like a required disclaimer or brand tone, need a custom evaluator.

The groundedness family

- **Groundedness:** is the response supported by the retrieved context? A **1 to 5** score from a judge model.
- **Groundedness Pro (preview):** a **pass or fail** from Azure AI Content Safety. **No model deployment** needed.
- **Ungrounded Attributes:** invented claims about a person, inferred from the conversation.
- **Response Completeness (preview):** did the answer miss something? Needs a reference answer.
- Groundedness checks against the context supplied. It can't catch a fact the context itself got wrong.

> ON THE EXAM
>
> Groundedness is precision: nothing added. Completeness is recall: nothing missed.

Groundedness needs the **context**. Similarity needs a **reference answer**. Relevance needs only the **question and response**.

AI-assisted evaluators or NLP metrics?

AI-ASSISTED

A judge model scores meaning

- Coherence, fluency, groundedness, relevance, similarity
- Needs a model deployment as the judge
- Costs tokens for every item scored
- Explains each score with a reason

NLP METRICS

Maths on overlapping words

- F1, BLEU, GLEU, ROUGE, METEOR
- Needs a reference answer
- Cheap, fast and deterministic
- Can't recognise the same meaning in different words

WATCH FOR THIS

A correct answer in different words scores badly on BLEU. That's the limit of word-overlap metrics.

Evaluator combinations by application

APPLICATION	COMBINE
RAG	Retrieval, Groundedness, Relevance, plus content safety
Agent	Tool Call Accuracy, Task Adherence, Intent Resolution, Rubric, content safety
Translation	BLEU, METEOR, Fluency, Coherence
Every application	Hate and Unfairness, Sexual, Violence, Self-Harm

ON THE EXAM

No single evaluator covers an application. Combine families, then add safety to all of them.

Agent evaluators

EVALUATOR	ASKS
Intent Resolution (preview)	Did it understand what the user wanted?
Task Adherence (preview)	Did it follow its instructions?
Task Completion (preview)	Did it finish the task end to end?
Tool Selection, Tool Input Accuracy	Right tool? Right parameters?
Tool Output Utilization, Tool Call Success	Used the results? Calls succeeded?
Customer Satisfaction (preview)	How satisfied would the user be overall?

ON THE EXAM

Tool Call Accuracy combines selection, parameters and efficiency. The narrower tool evaluators help pinpoint which part failed.

Rubric evaluators and Azure OpenAI graders

- **Rubric (preview):** your own weighted criteria, scored by a judge model.
- Returns a weighted average from 0 to 1, with reasoning for each criterion.
- **Azure OpenAI graders** cover common patterns without writing code:
- **Model Labeler** classifies; **Model Scorer** gives a numeric score; **String Checker** matches text; **Text Similarity** compares.

> ON THE EXAM
>
> Try a rubric or a grader before building a custom evaluator. They cover many 'our own criteria' requirements.

Three ways to build your own evaluator

TYPE	HOW IT SCORES	BEST FOR
Code-based	A Python grade() function, 0.0 to 1.0	Rules, keywords, format, length
Prompt-based	A judge prompt returning a result and a reason	Tone, subjective quality
Endpoint-based	Your own HTTP service returns the score	Network access, proprietary models

THE TIEBREAKER

Deterministic rule? Code-based. Needs judgement? Prompt-based. Needs your systems or network? Endpoint-based.

Limits of the grade() sandbox

- **No network access** while it runs. Calling an API from grade() won't work.
- Up to **2 minutes** per call, code under 256 KB, 2 GB of memory.
- Common packages are available: numpy, pandas, jsonschema, pydantic, nltk, rouge-score.
- An exception or timeout records **0.0** and marks the item as an error. Code defensively.
- Still requires a **deployment_name** and a **pass_threshold**, even though no model is called.

> THE SIGNAL
>
> Needing a database lookup during scoring rules out code-based. That is the endpoint-based evaluator's job.

A code-based compliance evaluator

```
def grade(sample: dict, item: dict) -> float:
    """Pass only if the disclaimer is present and the answer is short."""
    response = item.get("response", "")
    if not response:
        return 0.0
    has_disclaimer = "not financial advice" in response.lower()
    short_enough = len(response.split()) <= 120
    if has_disclaimer and short_enough:
        return 1.0
    if has_disclaimer or short_enough:
        return 0.5
    return 0.0
```

ON THE EXAM

Deterministic, cheap and repeatable: the same response always gets the same score.

Data sources for an evaluation run

SOURCE	WHAT HAPPENS	USE IT WHEN
Dataset (JSONL, CSV)	Scores existing responses	Outputs are precomputed
Model target	Queries a model, then scores	Comparing models or prompts
Agent target	Queries an agent, then scores	Testing prompt or hosted agents
Traces	Scores logged interactions	Real traffic, any framework
Synthetic (preview)	Generates queries, then runs	No test data yet

ON THE EXAM

Model router can be evaluated, but only as a target.

Building evaluation data you can trust

- Upload JSONL or CSV as a **versioned dataset**, reusable across runs and CI/CD pipelines.
- No data yet? **Generate synthetic** queries before launch, or when traffic is low.
- In production, **convert agent traces** into datasets from real conversations.
- Map fields to evaluators with **data_mapping**, such as {{item.query}}.
- Needs a **judge model deployment** for AI-assisted evaluators, and the **Foundry User** role.

> ON THE EXAM
>
> Synthetic data gets you started. Real traffic, converted from traces, keeps the test honest.

Score each turn, or the whole conversation?

TURN LEVEL (DEFAULT)

One response at a time

- Scores each individual response
- Most evaluators work here
- Finds a single bad answer

CONVERSATION LEVEL

The whole interaction

- Scores the complete multi-turn exchange
- Satisfaction, task completion, compliance
- Custom evaluators receive all messages

WATCH FOR THIS

Every evaluator in a run must support the level you choose. You can't mix levels in one run.

An evaluation workflow

01
Name the failure
What would hurt users most?

02
Build data
Dataset, synthetic, or traces.

03
Choose evaluators
Built-in, rubric, or custom.

04
Run and read
Scores and reasons, per item.

05
Compare runs
Before and after each change.

06
Gate and monitor
CI gate, then production.

ON THE EXAM

The MOT before launch, then the dashboard while driving: modules 05 and 08 continue this loop.

> A policy assistant uses RAG. Reviewers have found answers that quote clauses which don't exist in any retrieved document.
>
> The team wants to detect this automatically across 500 test questions. Nobody has written reference answers for them.

Which evaluator should the team use?

A. F1 score, comparing each response with a reference answer.

B. Fluency and Coherence, to catch answers that read badly.

C. Groundedness, checking each response against its context.

D. Relevance, checking each response addresses the question asked.

◆ PAUSE AND DECIDE

Which evaluator should the team use?

A F1 against references.

Needs reference answers, and nobody has written any.

B Fluency and Coherence.

Invented clauses can be beautifully written. These measure style, not truth.

C Groundedness against context.

Flags content not supported by the retrieved documents, with no references needed.

D Relevance.

An invented clause can be perfectly on topic. Relevance won't catch it.

THE SIGNAL

Invented content plus retrieved context plus no references: groundedness, by elimination and by design.

A wealth management assistant must include the exact phrase 'not financial advice' in every answer, and keep each answer to 120 words or fewer.

Compliance wants every test answer checked the same way every time, at the lowest possible cost.

What should the team build?

A. A code-based custom evaluator that checks the phrase and word count.

B. A prompt-based custom evaluator asking a judge model to check both rules.

C. A rubric evaluator weighting the disclaimer and length as two criteria.

D. The Groundedness Pro evaluator run against each answer's context.

◆ PAUSE AND DECIDE

What should the team build?

A Code-based evaluator.

Both rules are exact checks: deterministic, repeatable, and no judge tokens.

B Prompt-based evaluator.

A judge can miscount words and costs tokens per item. Not the same way every time.

C Rubric evaluator.

Also an LLM judge: flexible, but not deterministic for exact rules.

D Groundedness Pro.

Checks support in context, not phrases or length.

THE SIGNAL

'Exact phrase', 'word count' and 'the same way every time' are the vocabulary of code, not of a judge.

A retailer runs support agents built with LangChain, outside Foundry. They send OpenTelemetry traces to Application Insights.

The team wants to evaluate last week's real customer conversations for task adherence, without re-running them.

Which data source should the evaluation use?

A. An agent target, sending last week's queries to a Foundry agent.

B. Traces, scoring the interactions logged in Application Insights.

C. Synthetic data, generating new queries modelled on last week's topics.

D. A model target, sending the same queries to the underlying model.

◆ PAUSE AND DECIDE

Which data source should the evaluation use?

A Agent target.

Re-runs queries against a Foundry agent. These agents aren't in Foundry, and re-running is excluded.

B Traces.

Scores the real, already-captured interactions, including agents built outside Foundry.

C Synthetic data.

New, invented queries. The team wants last week's real conversations.

D Model target.

Generates new responses from a model, not the agent's actual behaviour.

THE SIGNAL

'Real past conversations' plus 'without re-running' plus 'built outside Foundry' is trace evaluation.

> A support agent's individual responses score well on relevance and fluency, turn by turn.
>
> Yet many customers leave multi-turn conversations with their problem unresolved. The team wants a metric for whole conversations.

What should the team do?

A Raise the pass thresholds on the existing turn-level evaluators.

B Add Fluency and Coherence to the existing turn-level evaluation run.

C Add Tool Call Accuracy to score each individual turn's tool usage.

D A conversation-level run with Task Completion and Satisfaction.

◆ PAUSE AND DECIDE

What should the team do?

A Raise turn thresholds.

Stricter per-turn scoring still never looks at whether the conversation resolved.

B Add Fluency and Coherence.

More turn-level writing metrics. Each turn already reads well.

C Add Tool Call Accuracy per turn.

Useful for diagnosis, but still judges steps, not the overall outcome.

D Conversation level, completion and satisfaction.

Scores the whole interaction's outcome, which is where the failure is.

THE TIEBREAKER

Good turns, bad outcomes: the failure only exists at conversation level, so that's where to measure.

Evaluation in five lines

- **Choose by the failure you fear.** Fabrication: groundedness. Off-topic: relevance.
- **Groundedness needs context; similarity needs references; relevance needs neither.**
- **Custom evaluators:** code for exact rules, prompts for judgement, endpoints for your systems.
- **Data:** datasets, model or agent targets, traces, or synthetic. Traces cover non-Foundry agents.
- **Turn or conversation level**, never both in one run.

ON THE EXAM

Next, module 16: observability, with tracing, token analytics, latency breakdowns and safety signals.